

**CATHOLIC JUNIOR COLLEGE**
JC2 PRELIM EXAMINATION
Higher 2**BIOLOGY****9744/02**
29 AUGUST 2022
2 HOURS

Additional Materials: writing papers

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid. DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

SECTION A] STRUCTURED QUESTIONSThere are **ten questions with multiple subparts** in Section A.Answer **all** questions in the spaces provided on the Question Paper.

For Examiner's Use	
Section A	100
1 [10]	
2 [10]	
3 [9]	
4 [10]	
5 [10]	
6 [10]	
7 [10]	
8 [10]	
9 [11]	
10 [10]	
Total	

SECTION A: STRUCTURED QUESTIONS

Answer **all** questions.

- 1 Fig. 1.1 shows part of an epithelial cell in a small intestine.

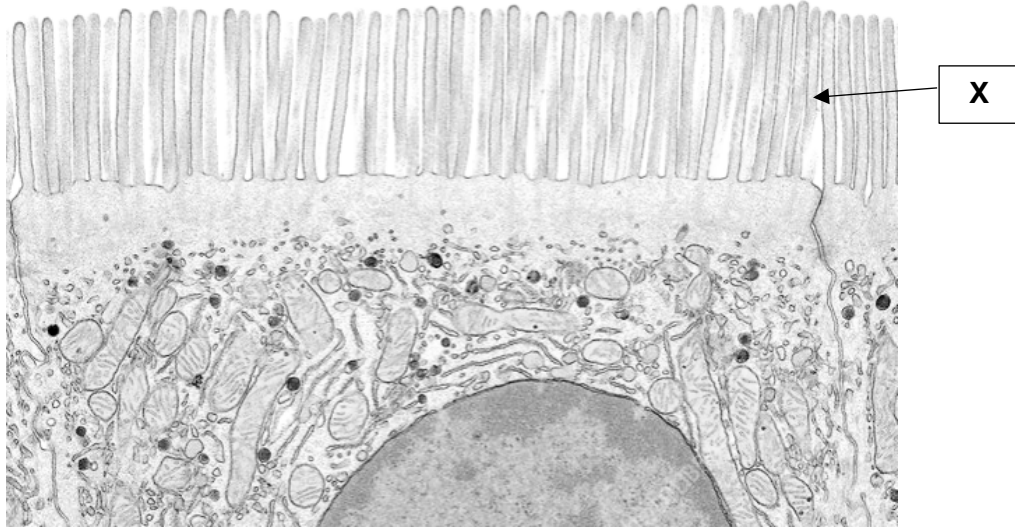


Fig. 1.1

- (a) Explain how structure X is adapted to perform its function.

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[2]

- (b) Explain why ions such as Na^+ and Cl^- cannot pass through structure X readily.

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[2]

Fig. 1.2 shows a sodium-potassium antiporter carrier protein, or Na-K protein pump, responsible in transporting ions in opposite directions across cell surface membrane of these epithelial cells.

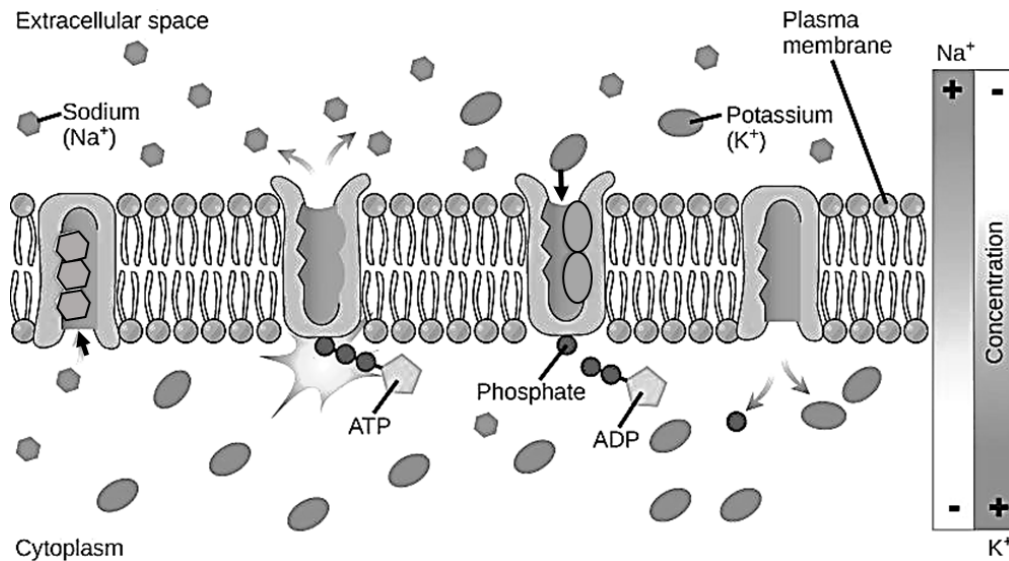


Fig. 1.2

- (c) Explain the type of amino acids required for the antiporter protein to perform its function.

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- (d) With reference to Fig. 1.2, suggest the significance of ATP in the functioning of Na-K protein pump antiporter.

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[Total: 10]

- 2 Taq polymerases are enzymes extracted from bacterium *Thermus aquaticus* which is often used in polymerase chain reaction (PCR). Fig. 2.1 shows the structure of Taq polymerase, which consist of one polypeptide chain.



Fig. 2.1

- (a) With reference to Fig. 2.1, describe the protein structure of taq polymerase.

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Fig. 2.2 below shows the effect of increasing temperature on the rate of reaction of the Taq polymerase.

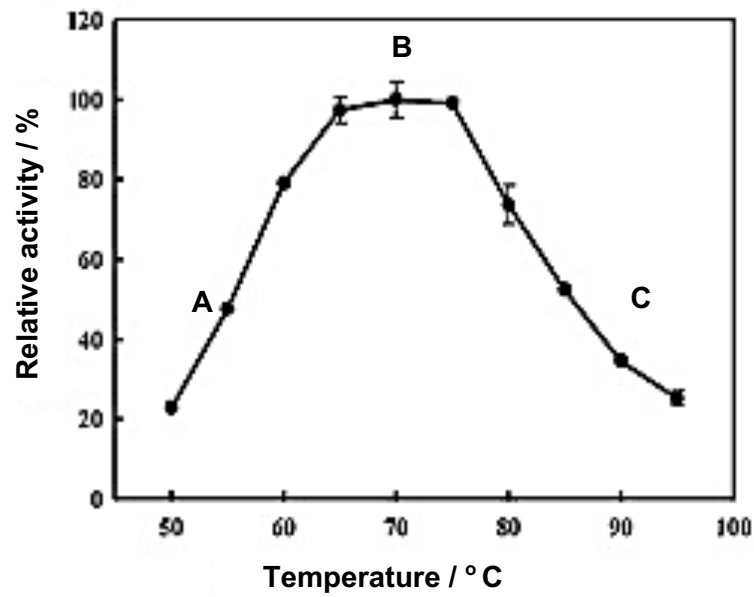


Fig. 2.2

- (b) With reference to Fig. 2.2, describe the curve between points A to C

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- (c) Taq polymerases are optimally active between 70°C to 75°C and is thermostable at high temperature. Suggest and explain one structural feature of taq polymerase that allow for its thermostability.

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- (d) The presence of metal ions in a biological sample can originate from various sources such as calcium (Ca^{2+}) from bone and iron (Fe^{3+}) in blood. The traces of such metal ions in samples for DNA analysis using PCR can have an inhibitory effect on taq polymerases on PCR amplification, thus reducing the efficiency of the amplification process. Ca^{2+} ions competitively bind to the polymerase whereas Fe^{3+} ions bind to the allosteric site of the enzyme.

Outline the differences in inhibition between Ca^{2+} ions and Fe^{3+} ions.

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[Total: 10]

- 3 Fig. 3.1 shows a ribosome in the process of translation.

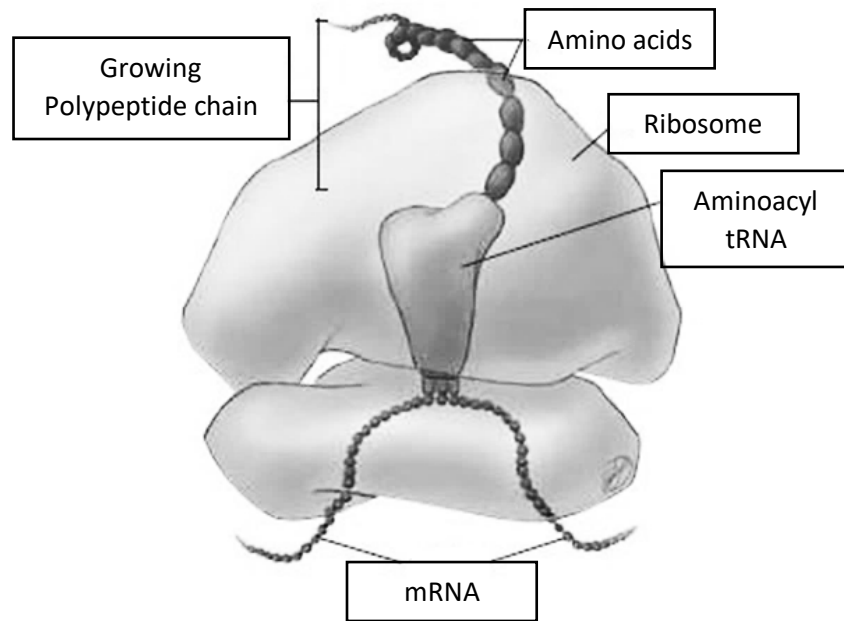


Fig. 3.1

- (a) With reference to Fig. 3.1 explain the **key features** leading to and during the process of translation that ensures that the information encoded on the mRNA results in the translation of a specific sequence of amino acids in the resulting polypeptide chain.

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- (b) Explain how tRNA maintains structural stability and why tRNA's 3D structure is important in its role in translation.

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- (ci) In the light of control of gene expression, suggest **and** explain two ways in which the presence of a protein may be prolonged within a eukaryotic body.

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- (cii) Gene control at the transcription level is sometimes affected by external stimuli. Suggest and explain how this is possible.

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[Total: 9]

- 4 Fig. 4.1 shows the structure of the *lac operon* in a prokaryote.

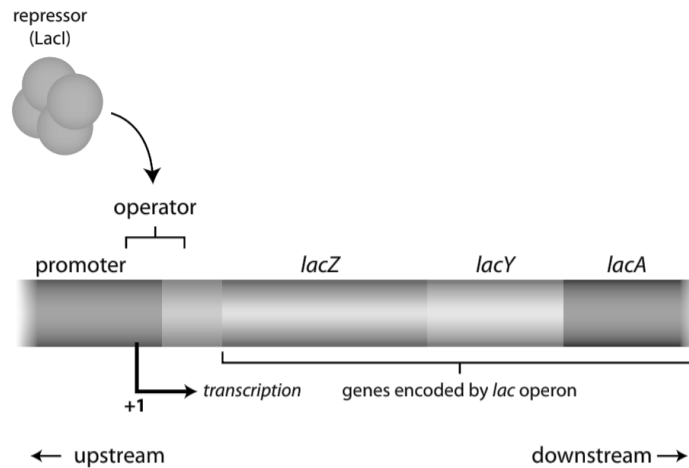


Fig. 4.1

- (a) With reference to Fig. 4.1, name two advantages of the use of Operons, explain your answer.

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Fig. 4.2 shows the rate of transcription of the lac operon in the presence of glucose, lactose and galactose (Graph A) and under the influence of the removal of a substance in Treatment 1 (Graph B).

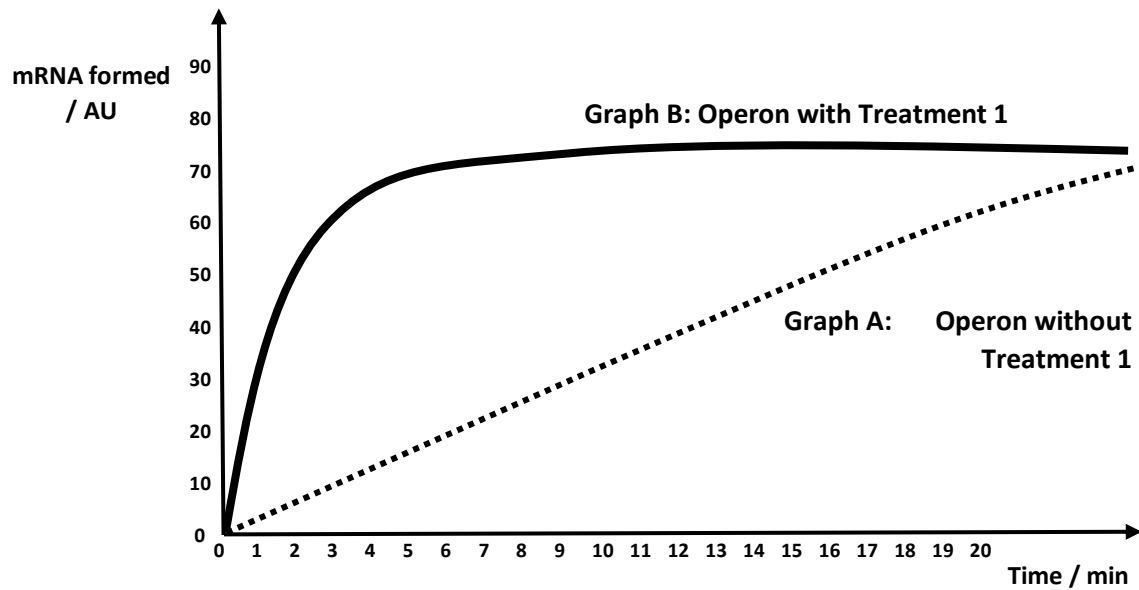


Fig. 4.2

With Ref to Fig. 4.2

(b) Explain what was removed in Treatment 1.

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(c) With reference to Fig. 4.2 Explain why transcription rate in Graph A is consistent throughout while in Graph B it decreases drastically to zero after 5min.

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- (d) There are different mechanisms by which genetic variation occurs in prokaryotes. Besides mutation, give two examples which result in lower genetic variation. Explain your answer.

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[Total: 10]

5 Mitosis is a key process in maintenance and growth of tissue in eukaryotes.

- (a) Explain how genetic stability is maintained through the process of Mitosis

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Cell division in is closely regulated by checkpoints within the cell cycle.

- (b) Explain the consequence of each of these checkpoints failing.

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- (c) In some cancers, the loss of function of the tumor suppressor gene e.g. p53, is a key contribution to the establishment of cancer. Excluding disease, explain how a loss of function of this gene might take place.

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- (d) Insurance premiums against critical illnesses like cancer increase significantly as the age at the point of application of the individual passes 50 years. Suggest why.

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[Total: 10]

- 6** Many genes affect a rabbit's fur colour, including one called tyrosinase (*TYR*). Animals homozygous dominant or heterozygous at a tyrosinase locus will produce coloured coats, while homozygous recessive rabbits develop unpigmented coats that appear white.

Fur colour is also partially established by a second gene called tyrosinase-related protein 1, or *TP1*. The dominant allele produces black fur, and the recessive allele produces brown or chocolate fur. Rabbits that inherit two recessive tyrosinase alleles will have white, unpigmented fur, regardless of which *TP1* alleles they inherit.

- (a) Explain how these two genes interact with each other to produce different colours of rabbits.

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- (b) A pet breeder was told that he could obtain brown rabbits by breeding a pure-bred white rabbit with a pure-bred black rabbit. However, he obtained a generation of all black rabbits. He then tried breeding one black rabbit from this generation with another white rabbit he bought from another source and obtained offspring in the following ratio.

Black	11
Brown	4
White	16

Assuming that the first generation of black rabbits were **heterozygous** at both gene loci, draw a genetic diagram to show how this ratio of offspring were obtained.

Use the following symbols for the respective genes:

T/t for *TYR*

P/p for *TP1*

- (c) Explain why it was not possible to obtain a brown rabbit by breeding any pure-bred black rabbits with white rabbits.

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.....[1]

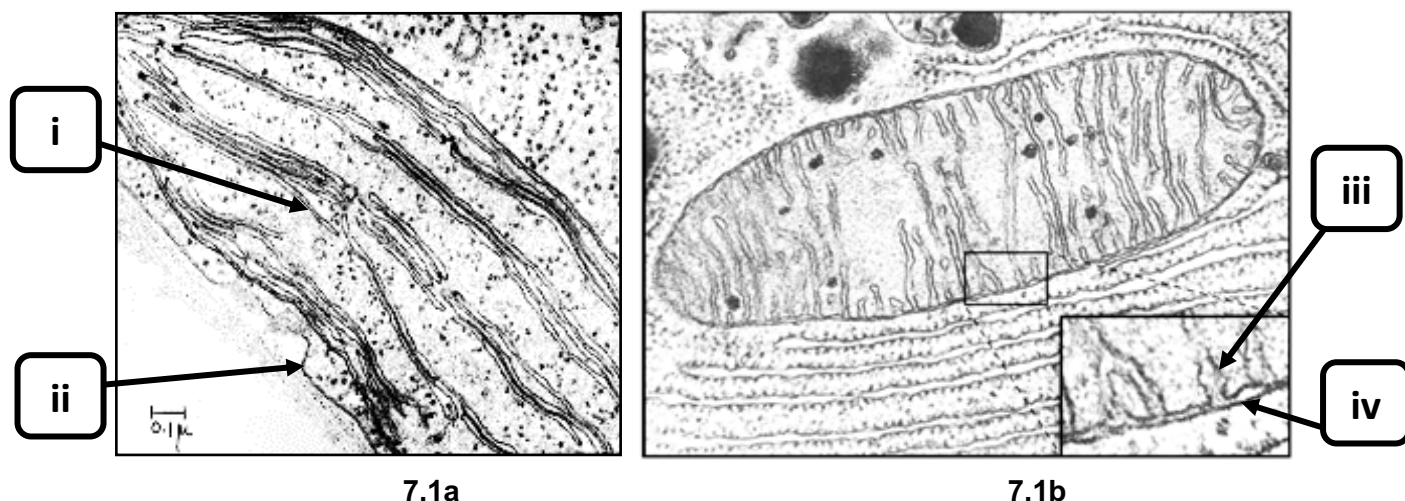
- (d) Himalayan rabbits are another species of rabbits which display different coat colours. However, their coat colours are known to be affected by temperature instead, having a generally white coat with their extremities known to display black colours due to cooler temperature.

With reference to the effect of tyrosinase, suggest how temperature could affect the phenotype of Himalayan rabbits with dominant alleles for the gene.

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[Total: 10]

7 Fig. 7.1a and 7.1b show the electronmicrographs of two organelles within a typical plant cell.



Both organelles use membranes as part of the process of generating ATP.

With reference to Fig 7.1a & 7.1b complete the information in the following table.

[3]

	7.1a	7.1b	
ai) Identify which membrane contains protein used directly in ATP synthesis.	*Circle one option i / ii	*Circle one option iii / iv	[1]
aii) Name 2 proteins are found on the membranes identified in (ai) associated with ATP production	Protein 1: Protein 2:		[2]

(b) An experiment was conducted using algal cells and the pH measurements were taken of the thylakoid space with results reflected as follows:

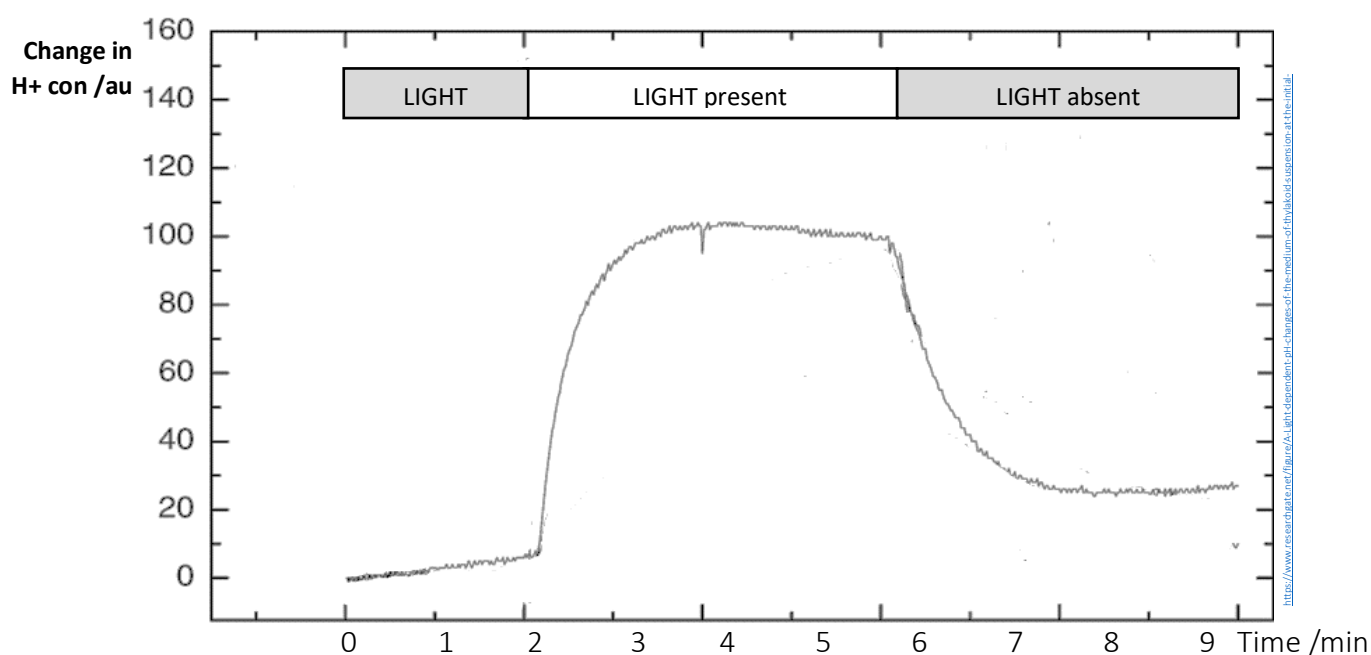


Fig. 7.2

- (i) With reference to Fig. 7.2, explain how light affects pH within the thylakoid membrane.

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- (ii) With reference to Fig. 7.2, suggest why there is still a change of 20 au in H^+ concentration even after light is absent from Time 6min onward.

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- (c) Upon illumination and in optimal conditions, glucose in the stroma is produced almost instantaneously seemingly without any lag. Explain how chloroplasts achieve this level of efficiency.

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[Total: 10]

- 8 The pancreas secretes both insulin and glucagon hormones into the bloodstream essential in regulating blood sugar levels.

(a) Outline the properties of these two hormones that allows them to function as ligands.

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Fig. 8.1 shows the signal transduction pathway in which adipose cells in response to insulin hormone.

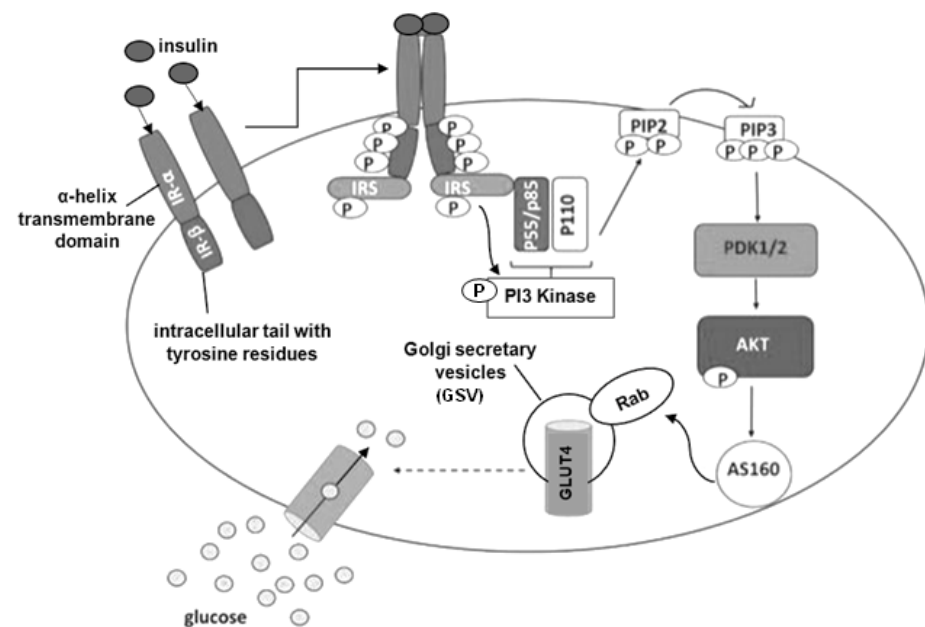


Fig. 8.1

- (b) With reference to Fig. 8.1,
- i. describe how autophosphorylation takes place

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- ii. suggest how an activated insulin receptor may bring about a decrease in the blood glucose concentration.

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- (e) Diabetes mellitus is a disorder in which the body does not produce enough or respond normally to insulin, causing blood glucose concentration levels to be abnormally high. Type II diabetes mellitus is characterised when target cells do not respond effectively to insulin hormone due to many factors, especially those individuals with a family history of Type II diabetes, being overweight and having sedentary and unhealthy lifestyle.

Suggest possible causes of why these target cells of such individuals become insensitive to insulin.

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[Total: 10]

- 9 Snakes are modern reptiles that appeared to have lost their limbs over time through a process known as regression evolution. There have been many hypotheses as to how this might have happened. They are thought to have evolved from either aquatic or burrowing ancestral lizards. Modern snakes include both aquatic and terrestrial species of which examples are shown below.

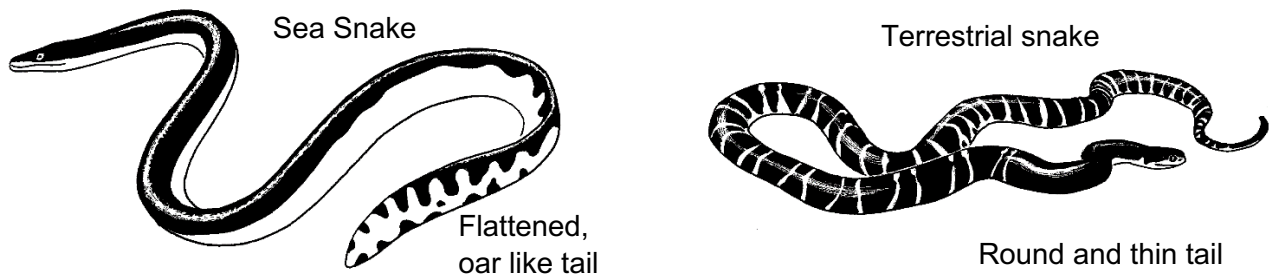


Fig. 9.1

- (a) With reference to the information above and Fig 9.1, suggest why the burrowing ancestral lizard is the more likely hypothesis.

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[1]

- (b) Researchers found that a series of mutations in the enhancer of a gene known as “Sonic hedgehog” disrupted the development of limbs in early snake embryos. This gene is responsible for the proper development of limbs in most animals. The gene was further found to have slightly higher expression in snakes that retained rudimentary limbs, such as pythons and boa constrictors, than completely limbless snakes, such as vipers and cobras, which showed even higher degradation of the enhancer sequences.

Explain how snakes could have evolved to become limbless.

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[4]

- (c) Despite the mutations of the enhancers of “Sonic hedgehog” gene, the gene remained relatively intact in most snakes. Furthermore, vestigial limbs such as those seen in Fig 9.2 continue to remain in some snakes.

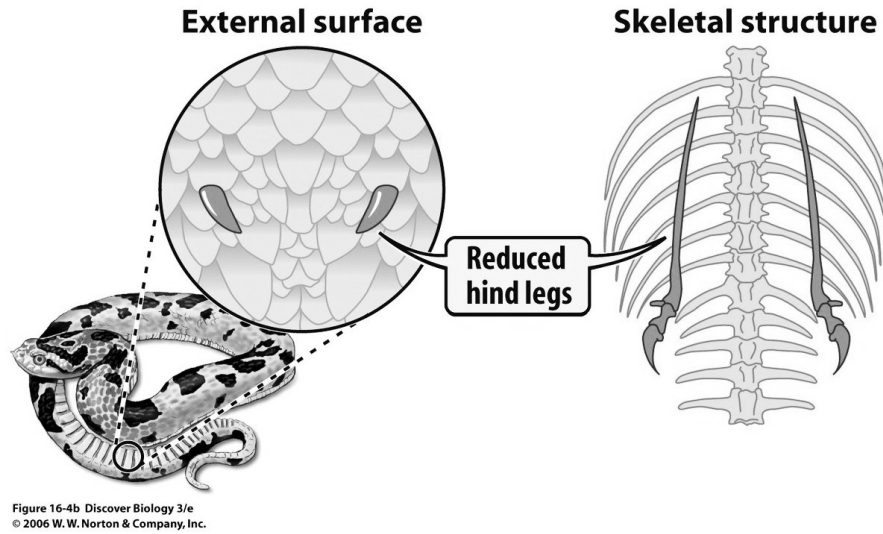


Fig. 9.2

- (i) Explain how the presence of “Sonic hedgehog” gene and vestigial limbs in snakes support the theory of evolution.

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- (ii) Explain how genes such as “Sonic hedgehog”, and vestigial organs such as those seen in Fig 9.2, could continue to remain in snakes.

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- (d) Fig 9.3 shows the phylogeny of snakes constructed from the analysis of molecular data from the genomes of different species of snakes.

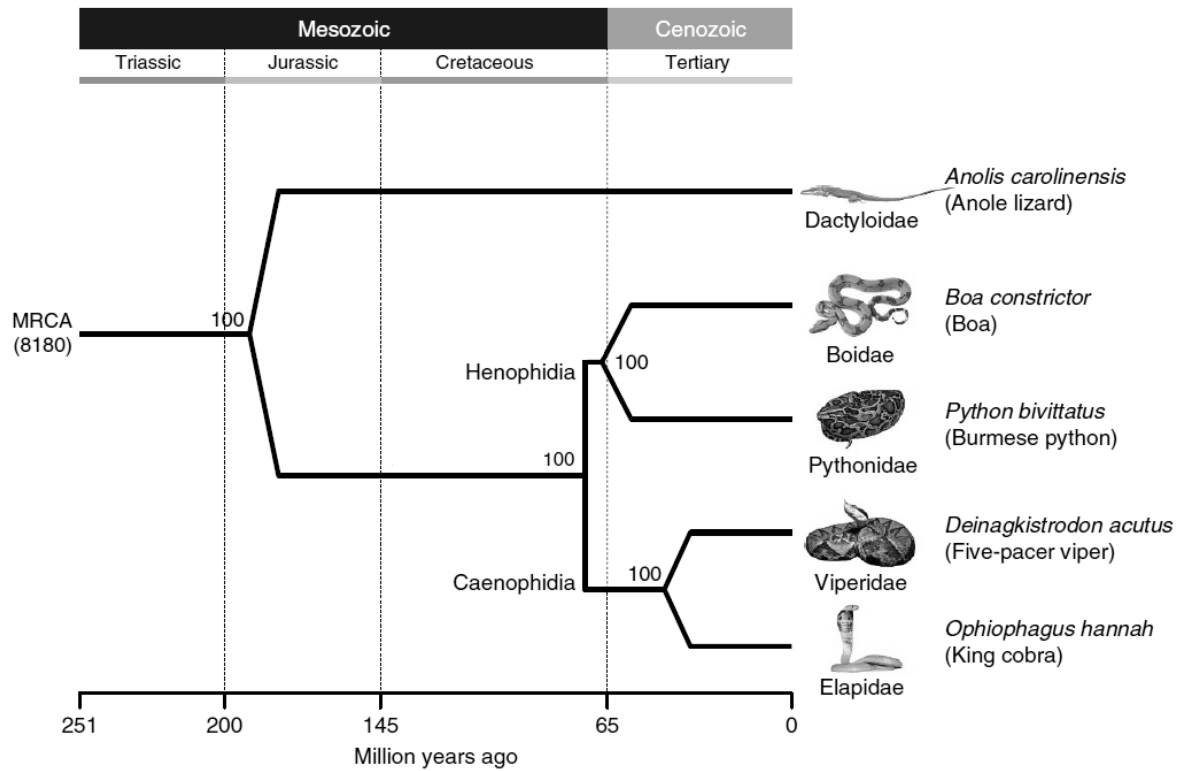


Fig. 9.3

Explain the advantages of molecular methods in determining phylogeny.

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[Total: 11]

- 10** The adaptive immune system in humans depends greatly on its ability to recognize the numerous pathogens which may in time invade the body.

Fig.10.1 shows the structure of the antigen receptor of a B cell

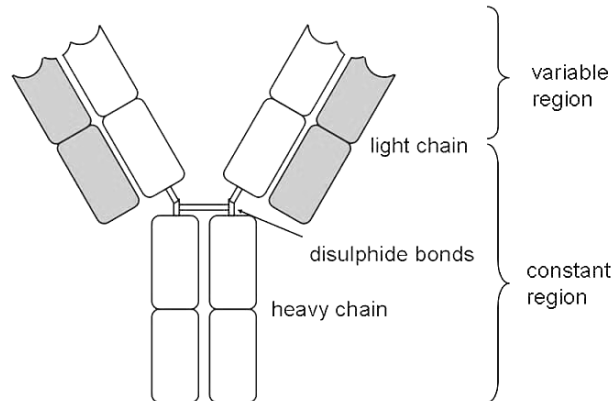


Fig. 10.1

With reference to Fig. 10.1

- (a)** In the light of recognition of potential pathogens, explain the advantage of having both light and heavy chains within the antigen receptor of the B Cell.

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In the fight against diseases, the process of vaccination helps to build resistance within a community against a given pathogen e.g. Influenza.

- (b)** In the light of effectiveness of vaccination against disease, suggest one advantage and one disadvantage in the use of this process.

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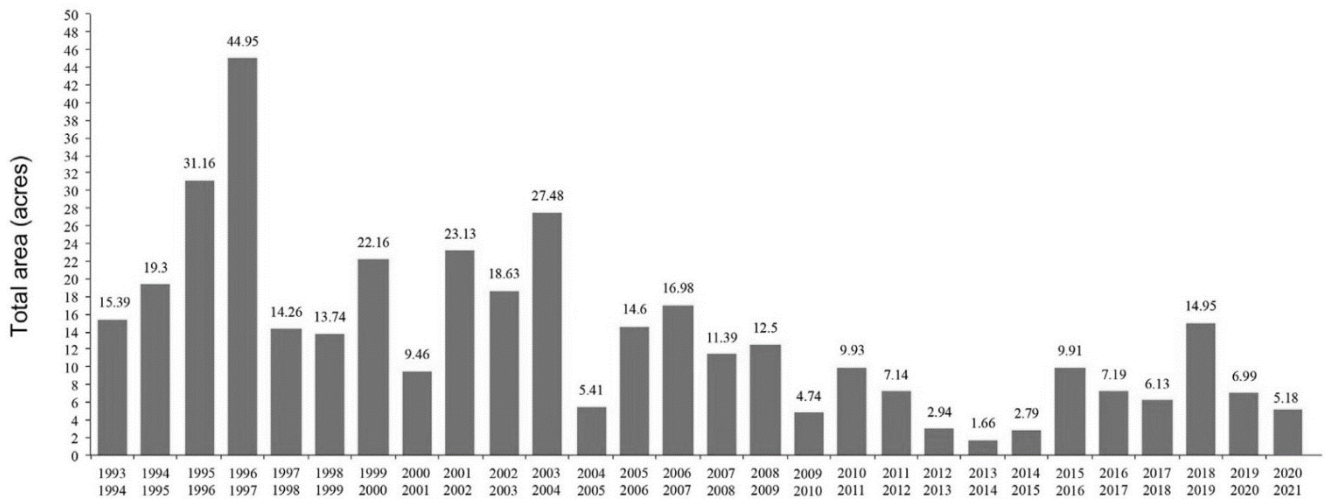
.....[2]

(c) Explain how a pathogen like Influenza causes disease in humans.

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[1]

Monarch butterflies, being one of the largest migrating butterflies in the world, recently come under the spotlight in research due to how global warming is affecting their populations. There was a reduction in their populations of about 26% compared to previous winter, when monarchs occupied only 2.83 hectares instead of 6 hectares, necessary to sustain the population.

Like most insects, they are highly sensitive to weather and climate which depend on changes in temperature to trigger reproduction, migration, and hibernation. Fig 10.2 below shows data of monarch butterflies hibernating in Mexico during winter.



Data collected from December to January every year

Press Release from WWF-Mexico 2020-2021

Fig. 10.2

(d) Explain why insects such as the monarch butterflies would be highly sensitive to climate change.

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[2]

Monarch butterflies feed on a specific milkweed plant species in which these plants secrete a toxin called cardenolides. Monarchs are usually resistant to these toxins which can be accumulated in their bodies to make them taste bad for their predators. However, based on recent research, it has been found that increasing temperature can cause milkweeds to secrete a significantly higher level of cardenolides than normal.

- (e) With reference to Fig. 10.2 and this information, suggest and explain how climate change is likely going to affect the monarch butterflies.

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[Total: 10]

END OF PAPER

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